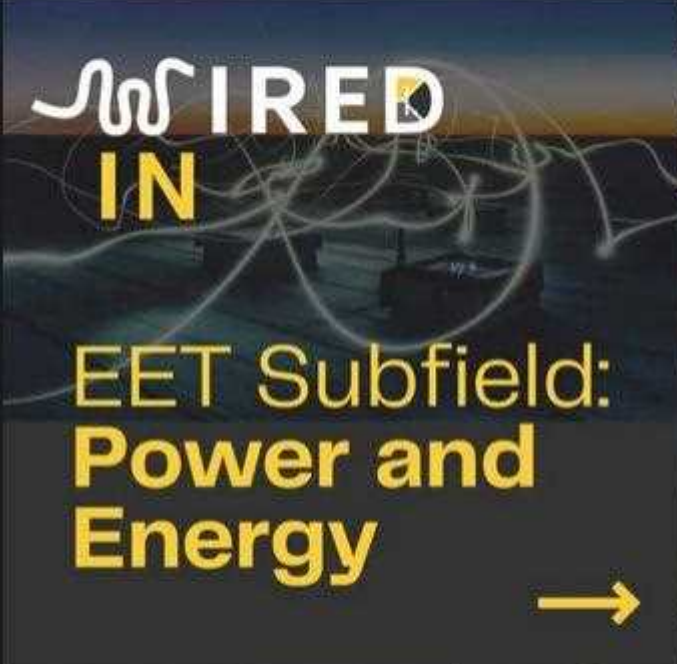


WIRED IN

Issue 02, 2026

**UNSW ELSOC**
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WIRED IN #2 ⚡ Ever thought about who actually keeps the grid running? Power & energy engineering goes way beyond power lines... think real-time problem solving, large-scale system design, and shaping how energy moves through the world. Swipe through to see what the industry actually looks like!



EET Subfield: Power and Energy





WHAT IS POWER AND ENERGY ENGINEERING?

Power Engineering deals with the generation, transmission, distribution and utilisation of electricity. Simply put, it is responsible for the design, construction and maintenance of all the systems that give us power!



WHAT DO POWER AND ENERGY ENGINEERS DO?

REAL-TIME OPERATION AND SUPPORT

Power Engineers must be able to oversee and provide support for the power systems both during normal operation and special circumstances. They must troubleshoot operational issues in real time, perform stability analyses for the grid during unusual conditions and approve orders for line crews to carry out.



WHAT DO POWER AND ENERGY ENGINEERS DO?

COMPLIANCE AND SAFETY

Safety is a top priority for all engineers! Power Engineers must continuously ensure that designs are compliant with Australian standards such as the AS 3000 (wiring), AS 4777 (inverters), AS 60076 (transformers) and many more. Safety and compliance reports must also be written to communicate the risks and hazards and show that designs are in line with the appropriate standards.



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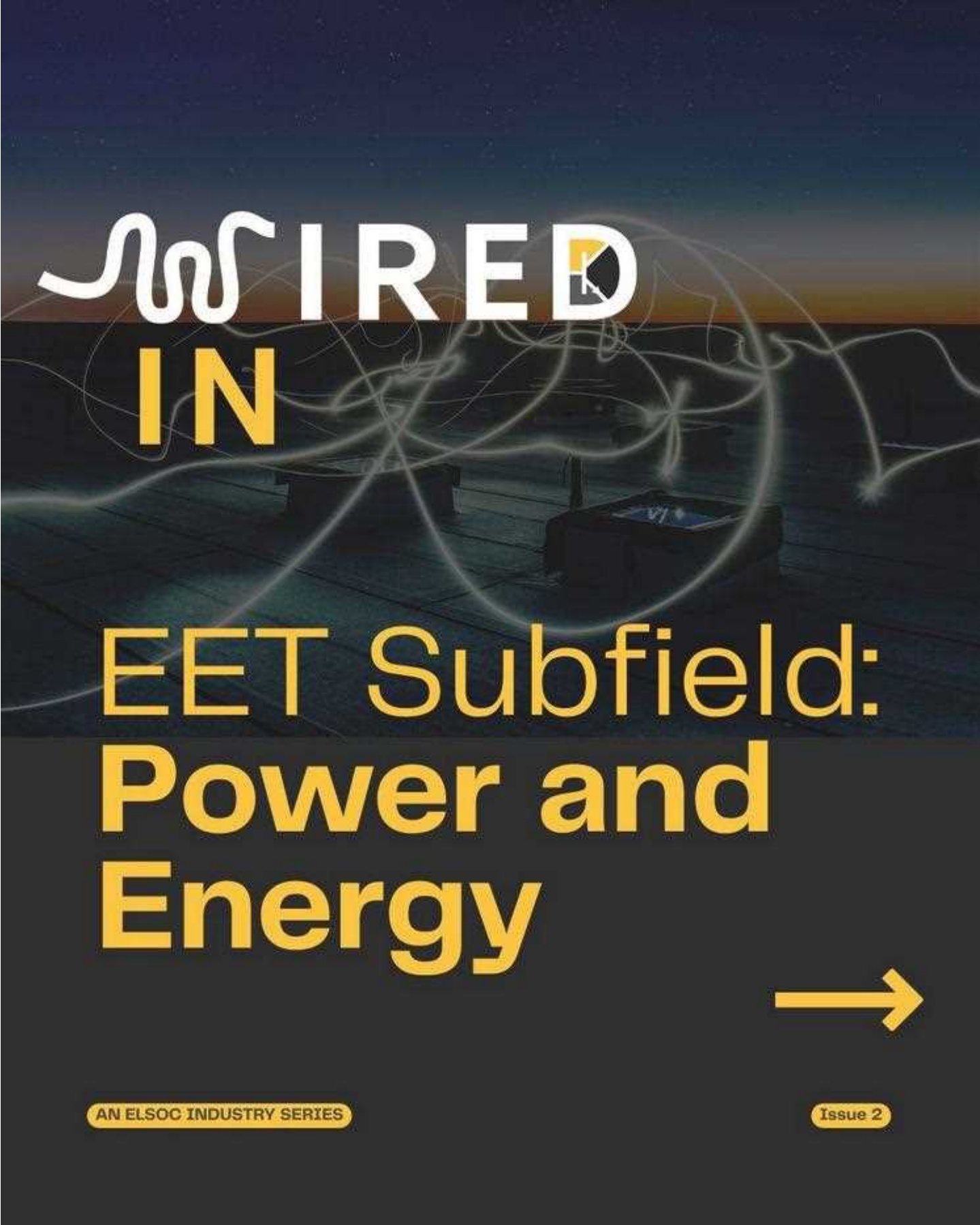
DESIGN AND MODELLING

Software such as ETAP (Electrical Transient Analyzer Program), PSS/E (Power System Simulation for Engineering) and PSCAD (Power Systems CAD) are used to model grid behaviour, to design and plan new electrical power systems.



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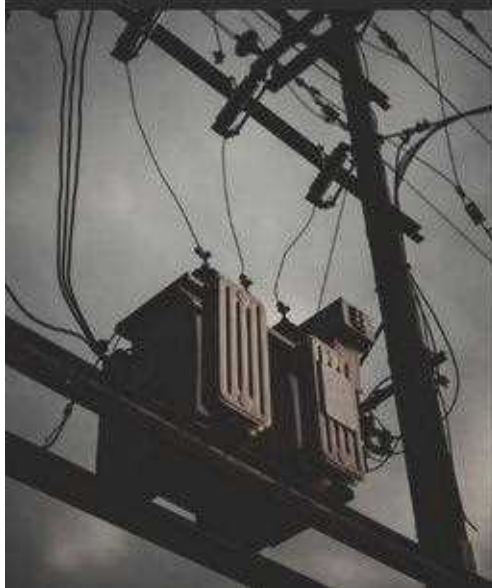


AN ELSOC INDUSTRY SERIES

Issue 2

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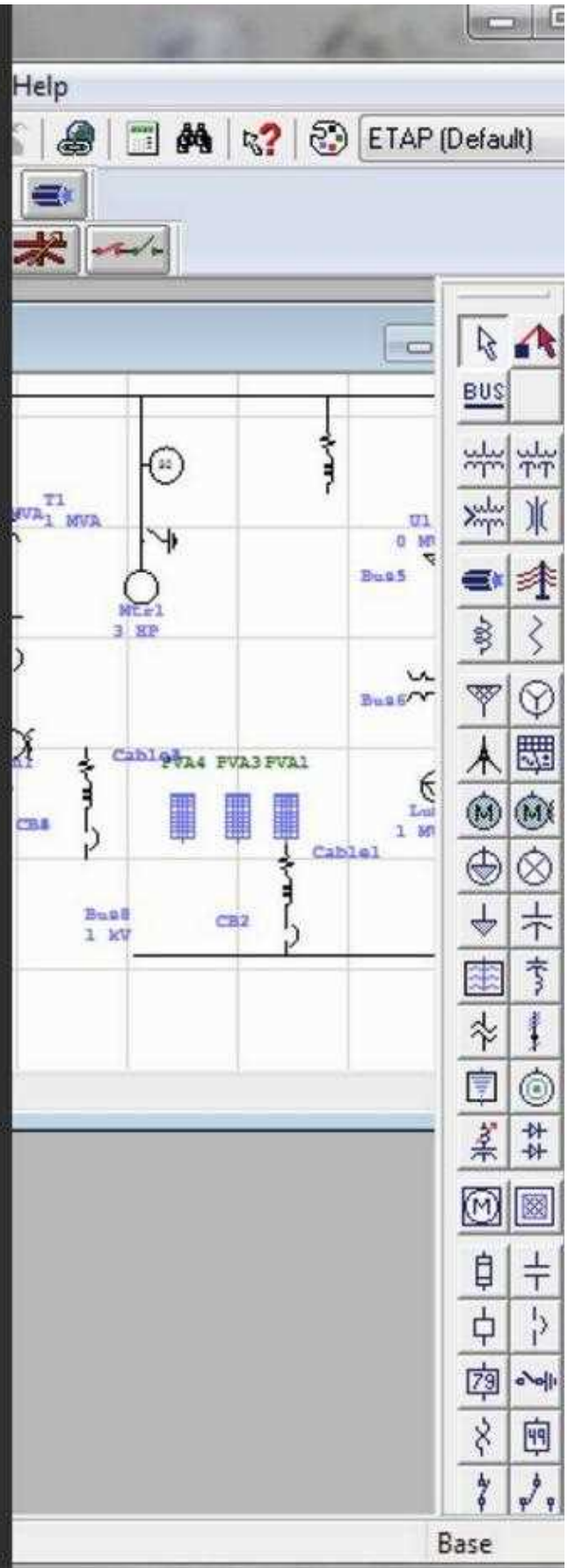
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WHAT DO POWER AND ENERGY ENGINEERS **NEED TO KNOW?**

Key areas in EET include:



Generation, distribution and transmission of electrical power (ELEC3105)



Circuit analysis fundamentals and AC power analysis (ELEC1111 + ELEC2134)



Power electronics (ELEC4614)



Simulation/modelling software (MATLAB, ETAP, PSS®E, etc.)



WHY SHOULD YOU PICK POWER AND ENERGY ENGINEERING?



Ability to create
more sustainable
future power
systems



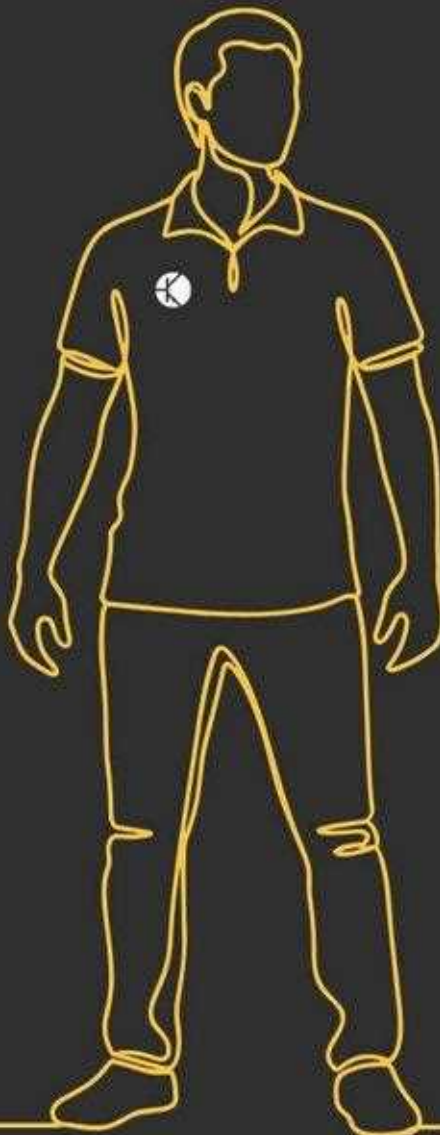
Exposure to
complex, large-
scale engineering
challenges



High demand and
job stability



High degree of real
world impact



KEY POWER AND ENERGY ENGINEERING **COMPANIES**



Transgrid (Power transmission)



Ausgrid, Endeavour Energy, Essential Energy (Power distribution)



AGL, EnergyAustralia, Origin Energy (Power generation)



Hitachi Energy, Siemens Energy, Schneider Electric
(Provide hardware for the grid)



Aurecon, Arup, Jacobs (Consultancy)

